

Deserialization of Untrusted Data in LiteDB - CVE-2022-23535

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Content

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Severity High

Score 9.8/10

Summary

LiteDB is a small, fast, and lightweight .NET NoSQL embedded database. In versions prior to 5.0.13 are subject to the Deserialization of Untrusted Data. LiteDB uses a special field in JSON documents to cast different types from `BsonDocument` to POCO classes. When instances of an object are not the same of class, `BsonMapper` use a special field `_type` string info with full class name with assembly to be loaded and fit into your model. If your end-user can send to your app a plain JSON string, deserialization can load an unsafe object to fit into your model. This issue is patched in version 5.0.13 with some basic fixes to avoid this, but is not 100% guaranteed when using `Object` type. The next major version will contain an allow-list to select what kind of Assembly can be loaded. Workarounds are detailed in the vendor advisory.

- Attack Complexity: LOW
- Attack Vector: NETWORK
- Integrity Impact: HIGH
- Scope: UNCHANGED
- User Interaction: NONE
- Privileges Required: NONE

- Confidentiality Impact: HIGH
- Availability Impact: HIGH

CWE-502 - Deserialization of Untrusted Data

Deserialization of untrusted data vulnerabilities enable an attacker to replace or manipulate a serialized object, replacing it with malicious data. When the object is deserialized at the victim's end the malicious data is able to compromise the victim's system. The exploit can be devastating, its impact may range from privilege escalation, broken access control, or denial of service attacks to allowing unauthorized access to the application's internal code and logic which can compromise the entire system.

References

- [Advisory](#)
- [Release Note](#)

Advisory Timeline

- Published Feb 24, 2023

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